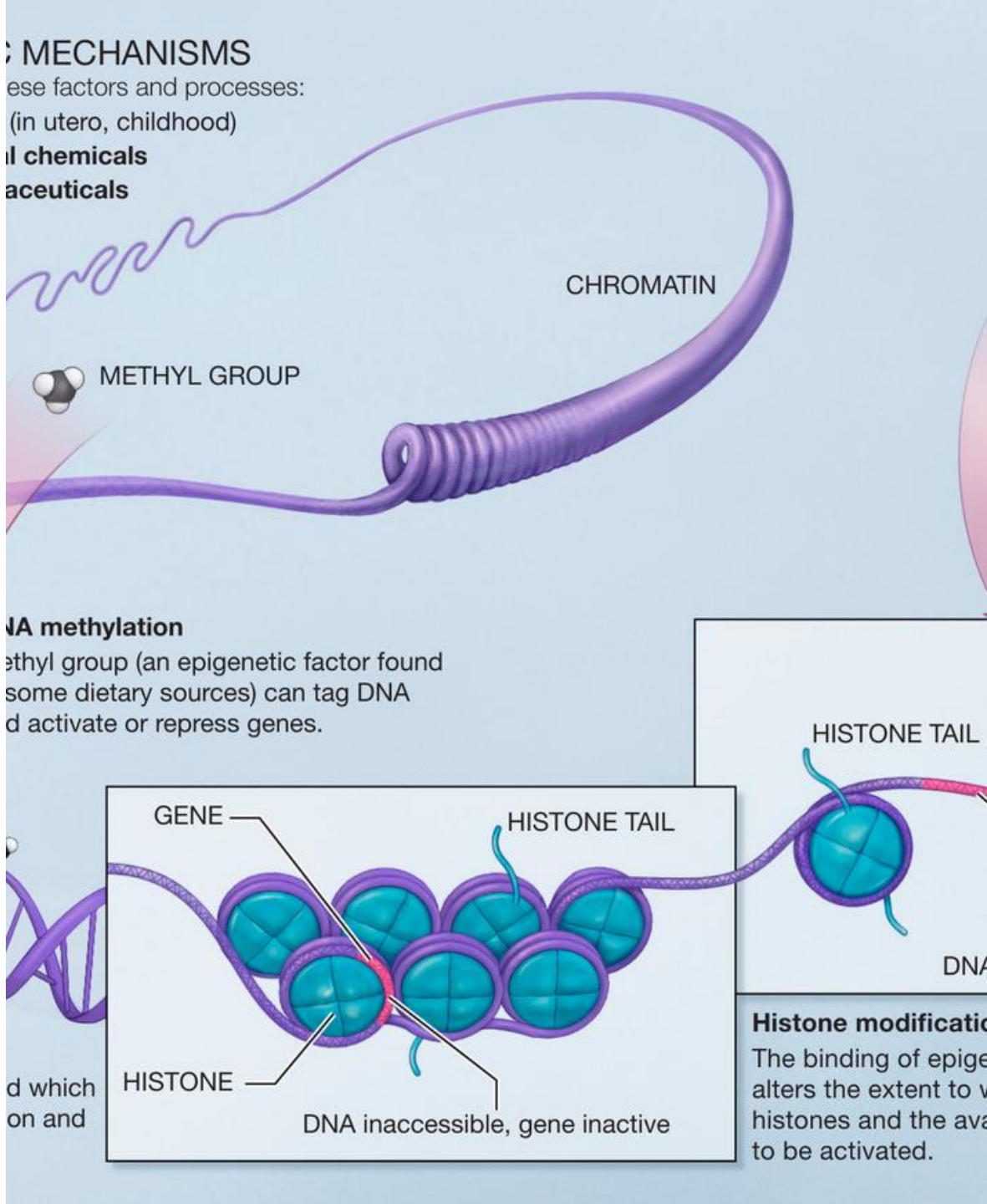
DP Biology – D2.2 (HL only)

Gene expression and epigenetics

- How can a cell change which genes it expresses?
- How are patterns of gene expression passed on?

HL statements D2.2.1–11. Textbook pp. 670–680.
Image: epigenetic mechanisms. National Institutes of Health, public domain.

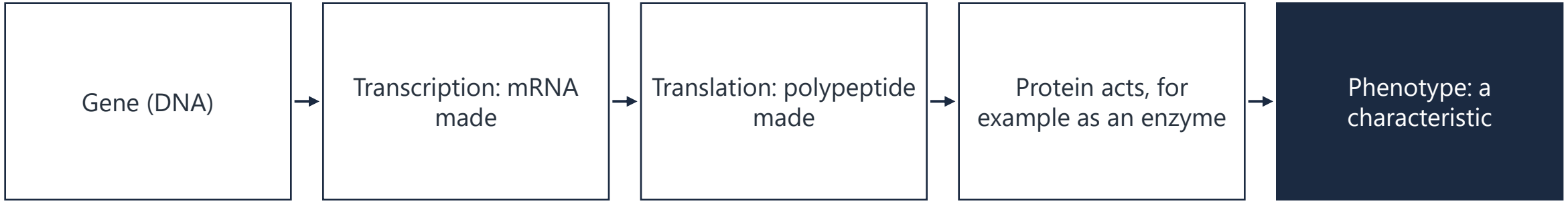


Objectives for each class

Class	Statements	By the end of the class, you can...
1	D2.2.1–3	Explain how genes affect the phenotype, and how transcription factors and mRNA breakdown control expression
2	D2.2.4–5	Explain epigenesis and compare genome, transcriptome and proteome
3	D2.2.6	Explain how methylation of DNA and histones acts as an epigenetic tag
4	D2.2.7–8	Explain epigenetic inheritance and give examples of the environment affecting gene expression
5	D2.2.9–10	Explain why most epigenetic tags are removed from gametes, using ligers and tigons, and evaluate twin studies
6	D2.2.11	Explain how a hormone and a nutrient in bacteria change patterns of gene expression

From gene to phenotype

D2.2.1



Example

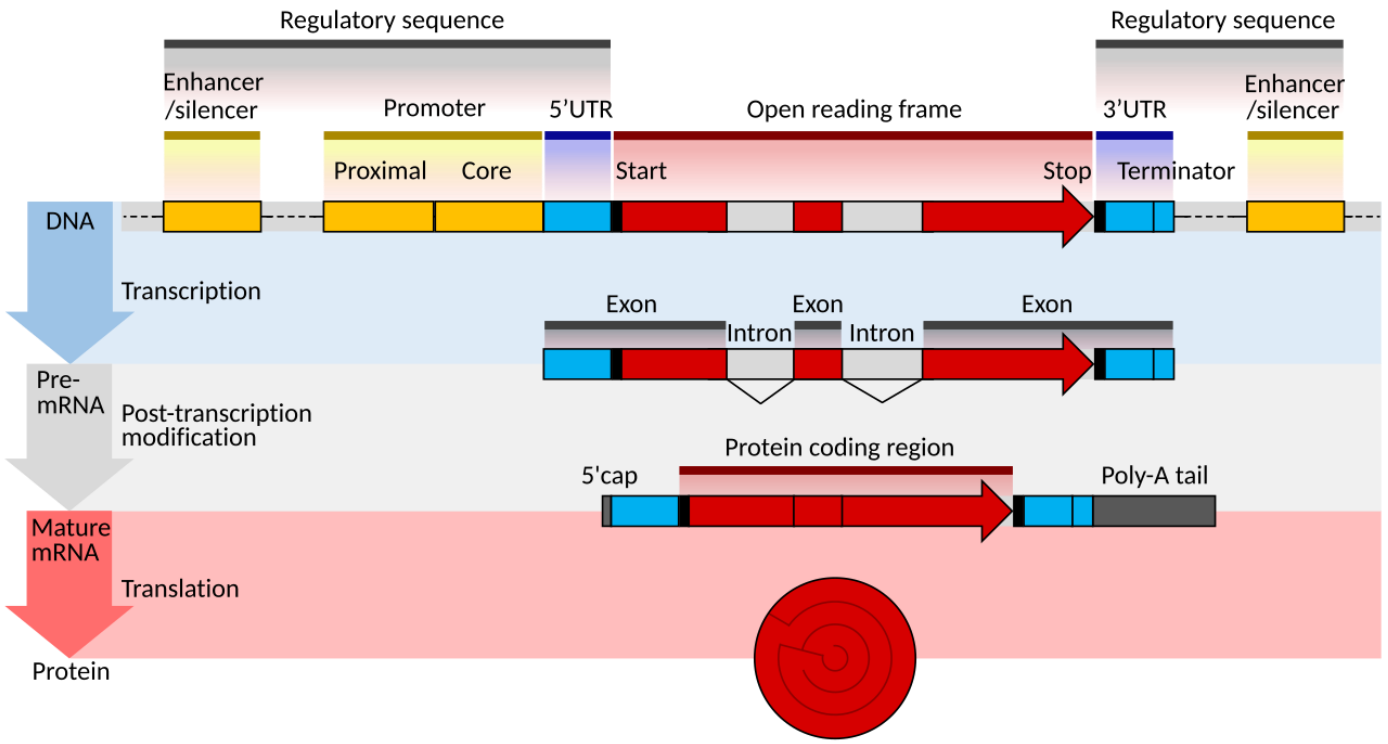
The gene for the enzyme tyrosinase is transcribed and translated in skin cells. The enzyme catalyses the first step in making the pigment melanin. A non-functional allele gives albinism. Each stage, from DNA to enzyme activity, is a point where expression can be controlled.

Key idea

Gene expression is the process by which the information in a gene has an effect on the phenotype. Most genes are expressed only in some cells, at some times.

Controlling transcription

D2.2.2



A eukaryotic gene with its regulatory sequences. Thomas Shafee, CC BY 4.0.

- **Promoter**: where RNA polymerase and general transcription factors bind, just before the gene
- **Enhancers**: sequences, sometimes thousands of bases away, that bind activator proteins and raise the rate
- **Silencers** bind repressors and lower it
- **Transcription factors** are proteins that bind specific base sequences

How distance works

DNA can loop, so an activator bound to a distant enhancer contacts the proteins at the promoter. Different cells contain different transcription factors, so they transcribe different genes.

Controlling mRNA breakdown

- After transcription, mRNA gains a 5′ cap and a poly-A tail at the 3′ end
- The tail is slowly shortened in the cytoplasm; when short enough, nucleases break the mRNA down
- In human cells mRNA lasts from minutes to days
- The longer an mRNA lasts, the more protein is made from it

Why break it down?

If mRNA lasted for ever, a cell could never stop making a protein. Short-lived mRNAs suit proteins needed briefly, such as signals; long-lived ones suit proteins needed in large amounts, such as haemoglobin in developing red blood cells.

Epigenesis: building a body

- **Epigenesis**: as a many-celled organism develops, its cells come to differ from one another
- Starting from one zygote, cells become different by switching genes on and off
- The DNA base sequence does not change, so the **genotype** is the same in all cells
- What changes is gene expression, and so the **phenotype** of each cell

Common mistake

Epigenetic changes are not mutations. A mutation changes the base sequence; an epigenetic change alters which genes are read.

Genome, transcriptome, proteome

D2.2.5

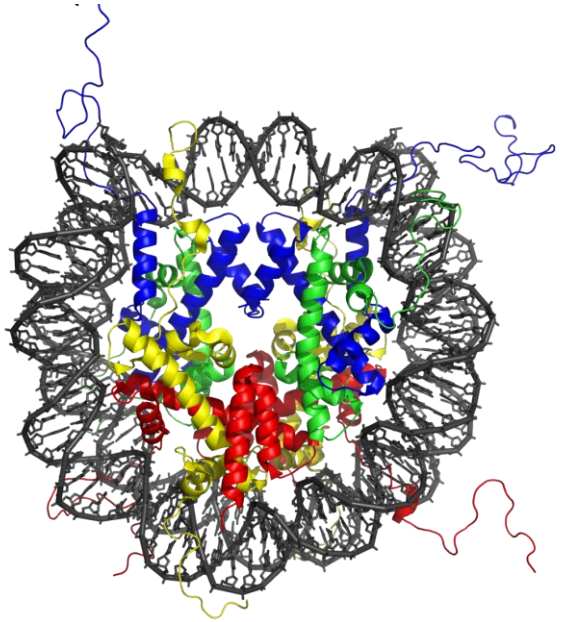
	What it is	Varies?
Genome	All the genetic information of an organism	The same in every cell of an individual
Transcriptome	All the mRNA made in a cell at a given time	Differs between cell types and changes with time and conditions
Proteome	All the proteins made in a cell at a given time	Differs between cells; depends on the transcriptome

Key idea

No cell expresses all of its genes. The pattern of genes expressed decides how the cell differentiates.

Epigenetic tags: methylation

D2.2.6



Nucleosome. Zephyris, CC BY-SA 3.0.

	DNA methylation	Histone methylation
Where	Cytosine in the promoter	Amino acids in histone tails
Effect	Represses transcription of the gene downstream	Can repress or activate transcription
How	Blocks transcription factors; DNA packs tightly	Changes how tightly DNA is packed and which proteins bind

Other tags

Acetylation of histones loosens packing and usually activates genes. Tags are added and removed by enzymes, so they can change during life. You do not need the details of how each tag works.

Passing on gene expression

D2.2.7

- When a cell divides by mitosis, methylation patterns are copied to the new DNA
- So daughter cells keep the same pattern: a liver cell gives rise to liver cells
- If tags survive meiosis and fertilization, a parent's changed expression can pass to offspring
- This is **epigenetic inheritance**: phenotype changes inherited without any change in base sequence

Dutch Hunger Winter

People whose mothers were pregnant during the famine of 1944–45 showed differences in methylation of some genes six decades later, and higher rates of some diseases. Discuss how strong this evidence is.

The environment changes gene expression

D2.2.8



Agouti mice. R. Jirtle and D. Dolinoy, CC BY 3.0.

- **Air pollution**: particulate matter, ozone and nitrogen oxides are linked to changes in DNA methylation, for example in genes for immune responses, with links to asthma
- **Diet**: in agouti mice, a mother's diet rich in methyl donors increases methylation of the agouti gene, giving brown, lean offspring
- **Stress, smoking, temperature** can also change tags

Evaluate

Many studies show correlation between exposure and methylation, not cause. Suggest what further evidence would show that a pollutant causes a change in methylation, and that the change causes disease.

Resetting tags in eggs and sperm

- Most tags are removed in the cells that become eggs and sperm, and again in the early embryo
- This resets the genome, so the zygote can develop into every cell type
- But some tags survive: in **imprinted** genes, the copy from one parent is silenced
- So whether an allele is expressed can depend on which parent it came from



A liger. Dinkun Chen, CC BY-SA 4.0.

Ligers and tigons

Liger: lion father, tiger mother; far larger than either. Tigon: tiger father, lion mother; similar size or smaller.

Why ligers are so big

D2.2.9

	Lion	Tiger
Mating	Females mate with several males	Females usually mate with one male
Imprinting in father	Growth-promoting genes active	Little growth promotion
Imprinting in mother	Growth-limiting genes active, to balance the father's	Little growth limitation needed

Put together

A liger gets growth-promoting tags from its lion father, but no matching growth-limiting tags from its tiger mother, so it grows very large. A tigon gets neither strong signal. The genes involved are not yet known: this is a hypothesis that fits the evidence.

Monozygotic twin studies

D2.2.10

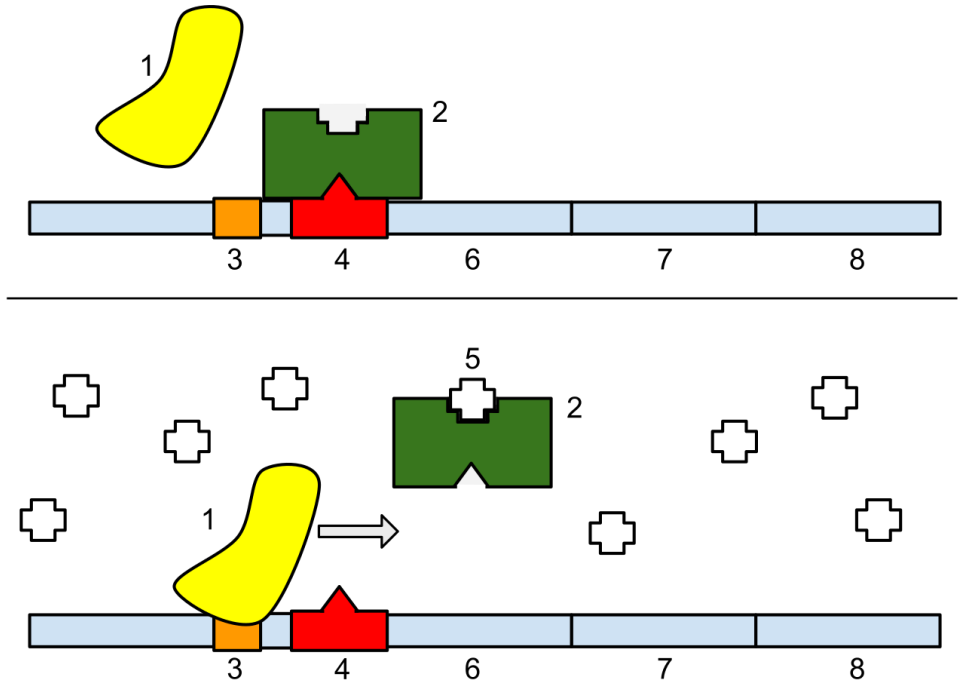
- Monozygotic (identical) twins come from one zygote, so they have the same genome
- Differences between them must come from the environment, including epigenetic changes
- Young twins have very similar methylation patterns; older twins differ more
- The more different their lifestyles, the more their patterns differ

Evaluate

Twins share a womb and often a home, so their environments overlap. Twins raised apart help separate the effects. Small samples and self-reported lifestyles are limitations.

Signals that switch genes

D2.2.11



Lac operon: lactose (white) removes the repressor (green). CC BY-SA 3.0.

Signal	Effect
Lactose in E. coli	Binds the repressor, which leaves the DNA, so genes for lactose uptake and breakdown are transcribed
Oestradiol (a hormone)	Enters target cells, binds a receptor, and the complex acts as a transcription factor, for example switching on the progesterone receptor gene

Why it makes sense

E. coli makes lactose-digesting enzymes only when lactose is present, saving energy and amino acids. A hormone lets one signal change gene expression in many target cells at once.

Check-in: gene expression

D2.2.1–11

1. Outline how transcription factors control which genes are transcribed.
2. Explain how the breakdown of mRNA helps control gene expression.
3. Compare the genome of a cell with its transcriptome and its proteome.
4. Explain how DNA methylation and histone modification affect transcription.
5. Outline why epigenetic tags are mostly removed in gametes.
6. Explain what monozygotic twin studies show about gene expression.

How to use this

Answer without your notes, then compare with a partner.

Working with data: methylation in twins

D2.2.10

Differences in DNA methylation between pairs of identical twins of different ages (relative units). These data are invented for practice; the pattern matches published studies.

Age of twins / years	Methylation difference
3	1.0
10	1.6
28	2.9
50	4.8

1. Describe the trend.
2. Suggest two reasons why identical twins become more different with age.
3. Explain why studying identical twins helps separate genetic and environmental effects.

Common mistakes to avoid

Not this	But this
Epigenetic changes alter the base sequence.	They change how genes are expressed, not the sequence.
Every cell expresses every gene.	Each cell type expresses a different set of genes.
Methylation always switches genes on.	Methylation of promoters usually switches genes off.
Identical twins are identical in every way.	Their genomes are the same, but epigenetic patterns diverge.

Check that you can...

1. **Explain** gene expression as the mechanism linking genotype and phenotype
2. **Explain** how transcription and mRNA breakdown are regulated
3. **Compare** a cell's genome with its transcriptome and proteome
4. **Explain** how methylation and histone modification act as epigenetic tags
5. **Outline** epigenetic inheritance and how environment affects gene expression
6. **Evaluate** evidence from monozygotic twin studies

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Distinguish between the genome and the proteome of a cell.
2. Explain how methylation of a promoter affects a gene.
3. Suggest why a liger grows larger than a tigon.

Image credits

Epigenetic mechanisms.png – National Institutes of Health – Public domain
Gene structure eukaryote 2 annotated.svg – Thomas Shafee – CC BY 4.0
Nucleosome 1KX5 colour coded.png – Zephyris at English Wikipedia – CC BY-SA 3.0
Agouti Mice.jpg – Randy Jirtle and Dana Dolinoy – CC BY 3.0
Liger – Shenzhen Safari Park 2023-02-03 01.jpg – Dinkun Chen – CC BY-SA 4.0
Lac Operon.svg – T A RAJU – CC BY-SA 3.0

All from Wikimedia Commons (commons.wikimedia.org, search the file name). Some images are cropped. CC licences: creativecommons.org/licenses